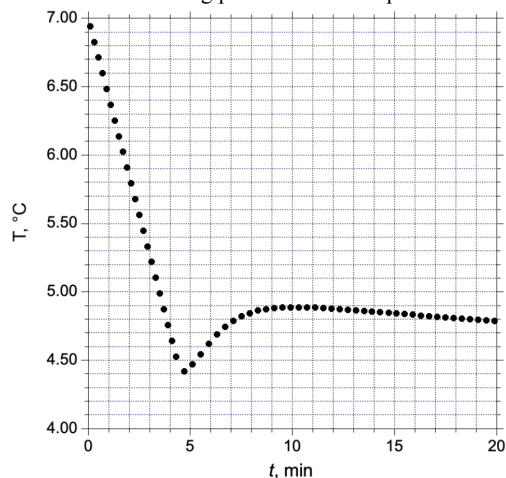


States Of Matter

1. H_2O has a higher normal boiling point than HF (100°C vs. 20°C). Which is the best explanation for this difference?
- (A) H_2O has a larger dipole moment than HF .
 (B) H_2O is more polarizable than HF .
 (C) H_2O forms stronger hydrogen bonds than HF .
 (D) H_2O forms more hydrogen bonds per mole than HF .

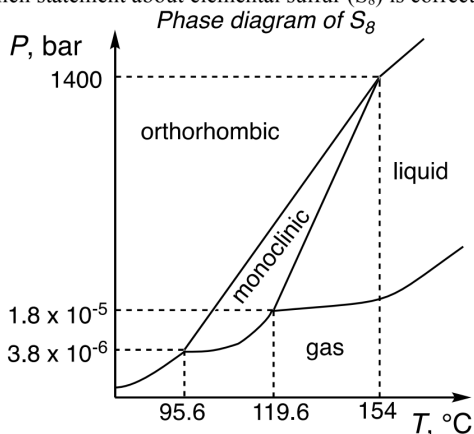
2. A solution of camphor in benzene is cooled slowly to determine its freezing point, resulting in the data shown below. What freezing point should be reported?



- (A) 4.38°C (B) 4.78°C
 (C) 4.88°C (D) 4.98°C
3. The vapor pressure of butane at 0°C is 1 atm. A 0.100 mol sample of liquid butane at 0°C is introduced into an evacuated, well-insulated 2.25 L container. Which statements correctly describe the system after it reaches equilibrium?
- I. The temperature is lower than 0°C .
 II. All of the butane is in the gas phase.
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
4. The triple point of krypton is at 0.73 bar and 116 K. Which is true of a pure sample of Kr at equilibrium at 0.73 bar and 90 K?
- (A) Only the gas phase is stable
 (B) Only the solid phase is stable
 (C) The solid and gas phases coexist
 (D) The solid and liquid phases coexist
5. Which statements about relative boiling points are correct?
- I. 2-Methylpropane has a higher normal boiling point than butane.
 II. 2-Methyl-1-propanol has a higher normal boiling point than 1-butanol.
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
6. Gold crystallizes in a face-centered cubic lattice with a density of 19.31 g cm^{-3} . What is the radius of a gold atom?
- (A) 144.1 pm (B) 176.5 pm
 (C) 203.8 pm (D) 288.2 pm
7. 8.82 g of Br_2 is placed in an evacuated 1.00 L flask and heated to 58.8°C , the normal boiling point of bromine. How much liquid bromine remains at equilibrium?
- (A) 0.00 g (B) 2.95 g
 (C) 3.63 g (D) 5.87 g
8. The normal boiling point of dichloromethane (CH_2Cl_2) is 40°C , while the normal boiling point of chloroform (CHCl_3) is 61°C . An increase in which type of intermolecular forces is most responsible for this increase in boiling point?
- (A) London dispersion forces
 (B) Dipole-dipole interactions
 (C) Ion-dipole interactions
 (D) Hydrogen bonding
9. Which has the highest melting point?
- (A) K_2O (B) CaO
 (C) Cu_2O (D) GeO_2
10. Platinum crystallizes in a face-centered cubic unit cell. The atomic radius of platinum is 139 pm. What is the density of solid platinum?
- (A) 5.33 g cm^{-3} (B) 7.54 g cm^{-3}
 (C) 18.6 g cm^{-3} (D) 21.3 g cm^{-3}

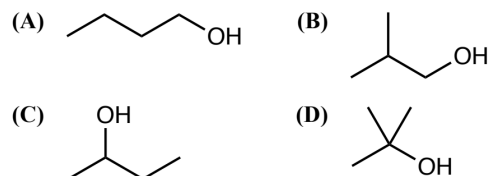
States Of Matter

11. Which statement about elemental sulfur (S_8) is correct?



- (A) Monoclinic sulfur sublimates at 1 bar pressure.
- (B) Monoclinic sulfur is denser than orthorhombic sulfur.
- (C) Monoclinic sulfur is not thermodynamically stable above 119.6 °C.
- (D) Monoclinic sulfur transforms exothermically into orthorhombic sulfur.
12. The unit cell of solid fluorite, CaF_2 , is shown below. Which statement about fluorite is correct?
-
- The diagram shows a cubic unit cell of fluorite (CaF_2). Calcium atoms (Ca) are represented by larger open circles and are located at the corners and the centers of the edges of the cube. Fluorine atoms (F) are represented by smaller filled circles and are located in the tetrahedral holes of the cube. There are 4 Ca atoms and 8 F atoms per unit cell.
- (A) Fluorite has a primitive cubic unit cell.
- (B) The larger open circles represent the Ca atoms and the smaller filled circles represent the F atoms.
- (C) The larger open circles' nearest neighbors consist of six of the other larger open circles arranged in an octahedron.
- (D) The smaller closed circles' nearest neighbors consist of eight of the larger open circles arranged in a cube.
13. A sample of pentane (bp = 36 °C) is in a rigid closed container at 50 °C and 1 atm. Which best describes what happens when the container is heated?
- (A) Liquid pentane vaporizes.
- (B) The intermolecular forces between pentane molecules become stronger.
- (C) The average kinetic energy of the pentane molecules increases.
- (D) The density of the sample decreases.

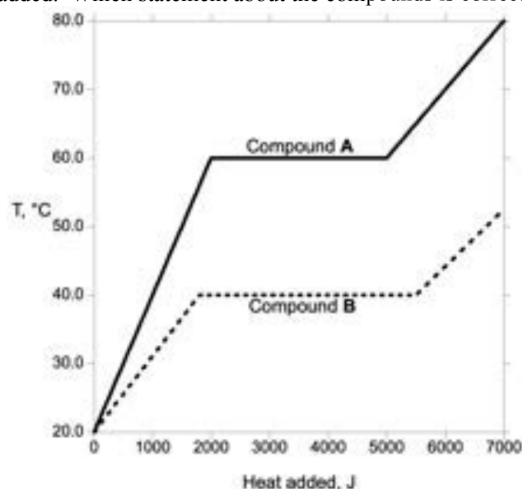
14. Which compound has the highest normal boiling point?



15. A mole of which gas has the smallest volume at 0 °C and 1 atm pressure?



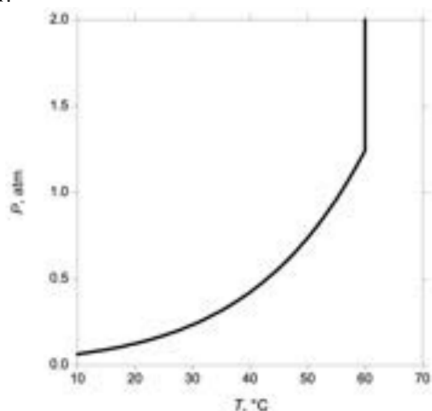
16. Two compounds, A and B, are both solids at 20 °C. In separate experiments, one mole of each is heated and its temperature measured as a function of the amount of heat added. Which statement about the compounds is correct?



- (A) The molar heat capacity of solid A is greater than that of solid B.
- (B) The molar heat capacity of liquid A is greater than that of liquid B.
- (C) The melting point of A is less than that of B.
- (D) The heat of fusion of A is less than that of B.

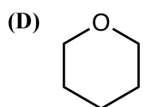
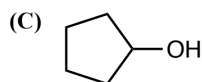
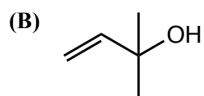
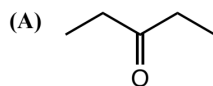
States Of Matter

17. A sample of a pure substance is placed in a sealed, rigid container and the pressure is measured as a function of temperature. Which is the best explanation for the result shown?



- (A) At lower temperatures, the substance is a mixture of solid and vapor, while at 60 °C the solid melts to give a mixture of liquid and vapor.
- (B) At lower temperatures, the substance is a mixture of liquid and vapor, while at 60 °C only liquid is present.
- (C) At lower temperatures, the substance is a mixture of liquid and vapor, while at 60 °C only a supercritical fluid is present.
- (D) At lower temperatures, the substance consists of vapor only, while at 60 °C only a supercritical fluid is present.
18. Cesium chloride ($M = 168.4$) crystallizes in a primitive cubic unit cell with each cesium ion surrounded by eight chloride ions arranged in a cube. The density of solid CsCl is 3.988 g cm^{-3} . What is the Cs–Cl distance?
- (A) 206.2 pm (B) 291.6 pm
- (C) 357.1 pm (D) 412.4 pm
19. A bicycle tire is filled with air to a pressure of 72 psi at a temperature of 25.0 °C. If the air temperature inside the tire increases to 55.0 °C, and the volume of the tire increases by 5.0%, what is the pressure inside the tire?
- (A) 69 psi (B) 75 psi
- (C) 83 psi (D) 166 psi

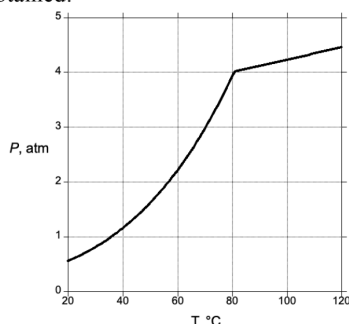
20. Which $\text{C}_5\text{H}_{10}\text{O}$ isomer has the highest normal boiling point?



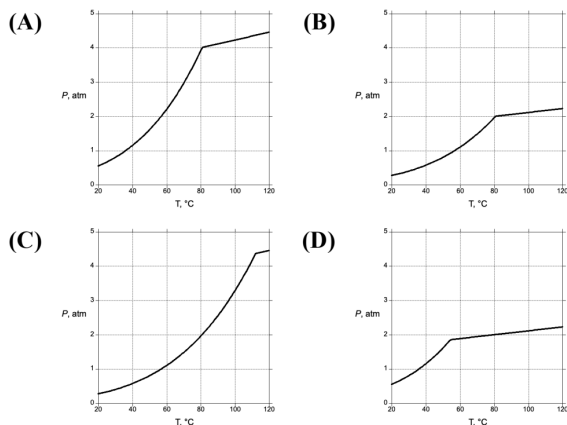
21. Two substances Q and R have the same normal boiling point. Substance Q has a larger enthalpy of vaporization than substance R. Which is true about the relative boiling points of the substances at 2.0 atm pressure?
- (A) The boiling point of Q is lower than the boiling point of R at 2.0 atm.
- (B) The boiling points of Q and R are the same at 2.0 atm.
- (C) The boiling point of Q is higher than the boiling point of R at 2.0 atm.
- (D) The relative boiling points cannot be predicted from the information given.
22. The density of white tin is 7.28 g cm^{-3} and the density of gray tin is 4.75 g cm^{-3} . The conversion of white tin to gray tin, $\text{Sn}(s, \text{white}) \rightarrow \text{Sn}(s, \text{gray})$, has a standard enthalpy of reaction of -2.1 kJ mol^{-1} at 298 K. At 1 atm pressure, white tin and gray tin are in equilibrium at 13 °C. Which best represents the phase diagram of tin near 13 °C and 1 atm?
- (A)
- (B)
- (C)
- (D)
23. Nickel crystallizes in a face-centered cubic structure. The edge of a unit cell has a length of 352 pm. What is the distance between the nuclei of two adjacent nickel atoms in this lattice?
- (A) 117 pm (B) 176 pm
- (C) 249 pm (D) 305 pm

States Of Matter

24. A 10-gram sample of a substance is added to a rigid evacuated 1 L vessel. The pressure in the vessel is measured as a function of temperature and the following graph is obtained:



Which graph is obtained if a 10-gram sample of the substance is heated in a 2 L vessel?



25. A 1.0-L container with 1.00 atm Ar gas is connected by a stopcock to a 2.0-L container containing 0.5 atm Kr gas. Which statement accurately describes the system after the stopcock is opened and the system is allowed to reach equilibrium? The temperature is 25 °C throughout the experiment.

- (A) The total pressure is 1.00 atm.
- (B) The number of moles of Ar in the 2.0-L container is equal to the number of moles of Kr in the 1.0-L container.
- (C) The average speed of the Ar atoms is equal to the average speed of the Kr atoms.
- (D) The partial pressure of Ar is equal to the partial pressure of Kr.

26. The normal boiling point of CH_2Br_2 is 97 °C while the normal boiling point of CHBr_3 is 149 °C. Differences in which interactions are most responsible for this difference in boiling points?

- (A) London dispersion forces
- (B) Dipole-dipole interactions
- (C) Hydrogen bonding
- (D) Carbon-bromine covalent bonding

27. The triple point of ethyne, C_2H_2 , is 192.4 K and 1.2 atm. What may be concluded from this information?

- (A) Solid ethyne sublimes rather than melting at 1 atm pressure.
- (B) Solid ethyne is less dense than liquid ethyne.
- (C) Solid ethyne is not thermodynamically stable at 193.0 K.
- (D) A sample of ethyne at 192.4 K must have a pressure of 1.2 atm.

28. The volume of a sample of gaseous SF_6 would increase by more than a factor of two under which of these conditions?

- I. It is heated from 300.0 K to 600.0 K at a pressure of 1.00 atm.
- II. The pressure is decreased from 2.00 atm to 1.00 atm at 300 K.

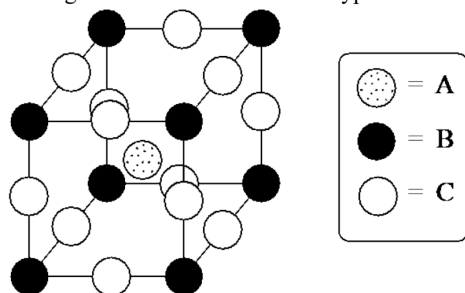
- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

29. A pure sample of benzene (C_6H_6) melts sharply at 5.5 °C, while a pure sample of hexafluorobenzene (C_6F_6) melts sharply at 5.2 °C. However, an equimolar mixture of C_6H_6 and C_6F_6 melts sharply at 23.7 °C. Which is the best explanation for this observation?

- (A) C_6H_6 and C_6F_6 are immiscible in the liquid phase.
- (B) C_6H_6 and C_6F_6 form a crystal containing equal amounts of the two substances.
- (C) Fluorocarbons characteristically show negative freezing point depression constants.
- (D) The enthalpy of mixing of C_6H_6 and C_6F_6 is negative.

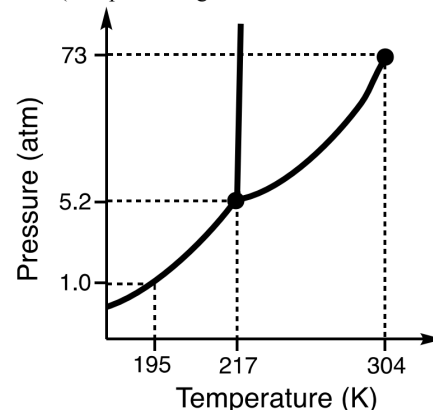
States Of Matter

30. A unit cell of a crystalline compound with the formula ABC_3 is shown below. What is the arrangement of nearest neighbors around the atoms of type C?



- (A) Linear arrangement of B atoms
(B) Square planar arrangement of A atoms
(C) Cubic arrangement of A atoms
(D) Cubic arrangement of C atoms
31. Which C_3H_6O isomer has the lowest normal boiling point?
- (A)
- (B)
- (C)
- (D)
32. Dichloromethane (CH_2Cl_2 , $M = 84.93$) is a volatile liquid with a vapor pressure of 0.53 atm at $24^\circ C$. 13.8 g of dichloromethane is introduced into a 5.0-L evacuated container maintained at $24^\circ C$. Which is the best description of the system at equilibrium?
- (A) One-third of the CH_2Cl_2 has evaporated.
(B) Two-thirds of the CH_2Cl_2 has evaporated.
(C) All of the CH_2Cl_2 has evaporated to give a pressure of 0.53 atm.
(D) All of the CH_2Cl_2 has evaporated to give a pressure of 0.79 atm.

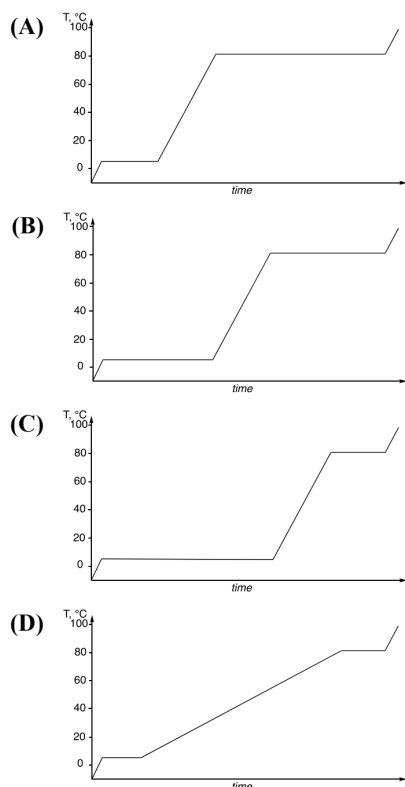
33. A piece of solid CO_2 initially at $-78^\circ C$ is placed in a small plastic container, which is then sealed and allowed to warm up in a room whose temperature is $22^\circ C$. Ultimately the plastic container explodes, ending any further observations. Which statement best describes the possibility of observing liquid CO_2 before the container explodes? (The phase diagram of CO_2 is shown below.)



- (A) Liquid CO_2 must be observable, since the explosion is driven by hydrostatic pressure of liquid CO_2 .
(B) Liquid CO_2 may be observed, since the triple point may be reached during this experiment.
(C) Liquid CO_2 may not be observed, since only sublimation of solid CO_2 is possible under the conditions of the experiment.
(D) Liquid CO_2 may not be observed, but supercritical CO_2 may be observed.
34. A solid calcium boride CaB_x has a density of 2.45 g cm^{-3} . It has a primitive cubic lattice with an edge length $a = 415 \text{ pm}$ and with one calcium atom per unit cell. What is the value of x ?
- (A) 2 (B) 4 (C) 6 (D) 8
35. The lattice enthalpy of Na_2O is $-2600 \text{ kJ mol}^{-1}$ while the lattice enthalpy of MgO is $-3900 \text{ kJ mol}^{-1}$. Which factors contribute to the larger magnitude of the lattice enthalpy of MgO ?
- I. Mg^{2+} has a larger charge than Na^+ .
II. Mg^{2+} is smaller than Na^+ .
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

States Of Matter

36. A sample of benzene (C_6H_6) is heated at a constant rate. Which graph most accurately represents how the temperature changes with time?



37. Which liquid has the highest vapor pressure at 25 °C?

- (A) CHCl_3 (B) CFCl_3
(C) CCl_4 (D) CBrCl_3

38. A sample of cyclohexane (C_6H_{12} , bp = 81 °C) is introduced into an evacuated container at 40 °C. One-third of the cyclohexane by mass is in the form of the liquid and two-thirds is in the form of the vapor. The volume of the container is then doubled, maintaining the temperature at 40 °C. What happens to the pressure as the volume is doubled?

- (A) The pressure remains the same.
(B) The pressure decreases by 25%.
(C) The pressure decreases by 33%.
(D) The pressure decreases by a factor of two.

39. The normal boiling points of ammonia, water, and hydrogen fluoride increase in the order NH_3 (bp = -33 °C) < HF (bp = 19 °C) < H_2O (bp = 100 °C). What is the best explanation for this ordering?

- (A) H_2O forms a greater number of hydrogen bonds per mol than NH_3 or HF.
(B) H_2O forms stronger hydrogen bonds than NH_3 or HF.
(C) H_2O has a larger dipole moment than NH_3 or HF.
(D) H_2O has stronger London dispersion forces than NH_3 or HF.

40. Which gas has the smallest molar volume at STP?

- (A) He (B) NH_3 (C) N_2 (D) CO

41. Which solid has the highest melting point?

- (A) OF_2 (B) MgF_2 (C) SiO_2 (D) ClO_2

42. A portion of the structure of solid potassium is shown below. In what type of unit cell are the atoms arranged?



- (A) Primitive cubic
(B) Body-centered cubic
(C) Face-centered cubic
(D) Hexagonal closest packed

43. Which set of characteristics describes a metal?

- (A) Melting point 180 °C, soft, electrically conductive as a solid
(B) Melting point 388 °C, soluble in CCl_4
(C) Melting point 801 °C, dissolves in water to give an electrically conducting solution
(D) Melting point 1400 °C, insoluble in water, has electrical conductivity that is low at room temperature but rises sharply as temperature increases

States Of Matter

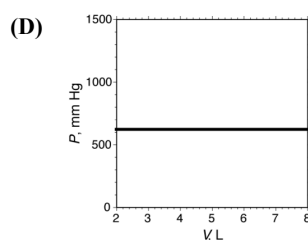
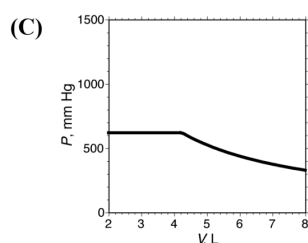
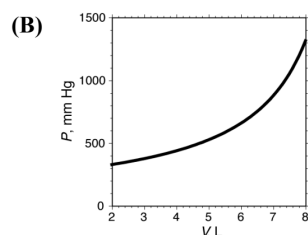
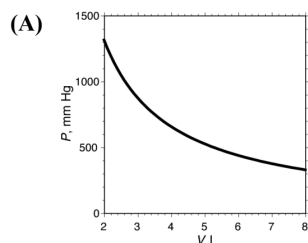
44. A test tube is filled with 5 mL water, 5 mL carbon tetrachloride (CCl_4), and a few crystals of iodine (I_2). After the tube is capped and shaken, what is the final appearance of the system after it is allowed to stand for a few minutes?

(A) A single layer of violet liquid.
 (B) A single layer of colorless liquid under violet vapor.
 (C) A colorless liquid layer above a violet liquid layer.
 (D) A violet liquid layer above a colorless liquid layer.

45. Which substance has the highest normal boiling point?

(A) CH_2CH_2 (B) NH_2NH_2
 (C) CH_3NH_2 (D) CH_3OH

46. An adjustable-volume container holds 10.0 g pentane ($M = 72.15$, normal boiling point = 36.1°C). The pressure in the container is measured as a function of volume while maintaining the temperature at 30°C . Which graph shows the results of this experiment?



47. Cotton balls soaked in concentrated solutions of HBr and CH_3NH_2 are placed at the same time in opposite ends of a glass tube. What is observed?

(A) A white ring of $(\text{CH}_3\text{NH}_3)\text{Br}$ forms in the tube closer to the side with the HBr .
 (B) A white ring of $(\text{CH}_3\text{NH}_3)\text{Br}$ forms in the tube closer to the side with the CH_3NH_2 .
 (C) A white ring of NH_4Br forms in the tube closer to the side with the HBr .
 (D) A white ring of NH_4Br forms in the tube closer to the side with the CH_3NH_2 .

48. A metal with a face-centered cubic (fcc) lattice has a unit cell edge length $a = 380.3 \text{ pm}$ and a density of 12.45 g cm^{-3} . What metal is it?

(A) K ($Z = 19$) (B) V ($Z = 23$)
 (C) Rh ($Z = 45$) (D) Pb ($Z = 82$)

49. A substance has a density of 2.38 g mL^{-1} and is not significantly compressible. The substance is most likely in which physical state?

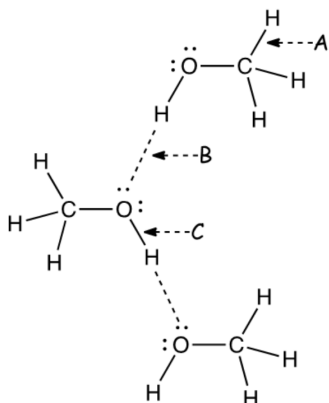
(A) Solid
 (B) Liquid
 (C) Gas
 (D) Solid or liquid; there is not enough information given to distinguish between the two

50. Carbon tetrachloride, CCl_4 , has a higher normal boiling point than chloroform, CHCl_3 (77°C vs. 61°C). Differences in which interactions contribute the most to this difference in boiling point?

(A) Hydrogen bonding
 (B) Dipole-dipole interactions
 (C) London dispersion forces
 (D) Covalent bonding

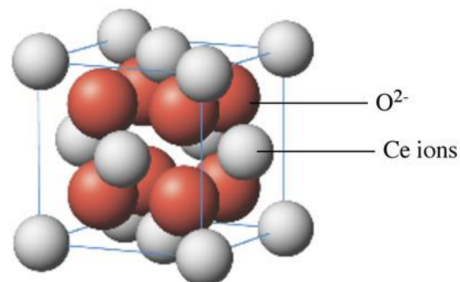
States Of Matter

51. The diagram represents a molecular view of a sample of liquid methanol, CH_3OH . Which letters in the diagram show hydrogen bonds?

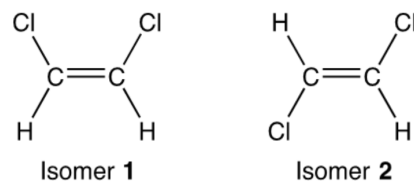


- (A) A only (B) B only
(C) C only (D) A and C
52. The triple point of ammonia is 195.40 K at 0.006 bar, while its melting point is 195.42 K at 1 bar pressure. Which statement about ammonia is correct?
- (A) Solid ammonia is denser than liquid ammonia at 195.4 K.
(B) Ammonia vapor cannot coexist with liquid ammonia at equilibrium at 195.5 K.
(C) Ammonia vapor cannot coexist with solid ammonia at equilibrium at 195.3 K.
(D) Solid ammonia and liquid ammonia cannot coexist at equilibrium at 195.5 K.
53. Acetone, $\text{C}_3\text{H}_6\text{O}$, has a vapor pressure of 0.307 bar at 25 °C. A sample of 0.100 mol acetone is added to a container that contains 1.00 L of argon gas at 1.00 bar pressure and 25 °C. The volume of the container is then increased to 4.00 L while maintaining the same temperature. What is the pressure in the container after the expansion?
- (A) 0.250 bar (B) 0.307 bar
(C) 0.557 bar (D) 0.870 bar

54. What is the formula of the oxide whose unit cell is shown?



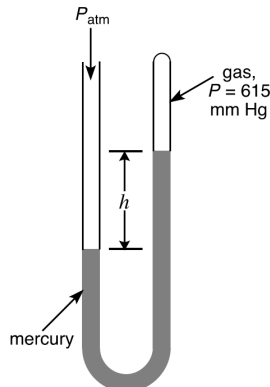
- (A) Ce_7O_4 (B) CeO (C) Ce_2O_3 (D) CeO_2
55. Which statement best describes the results of attempted separation by fractional distillation of the two isomers of 1,2-dichloroethene?



- (A) The two isomers can be separated by fractional distillation, with isomer 1 boiling at the lower temperature.
(B) The two isomers can be separated by fractional distillation, with isomer 2 boiling at the lower temperature.
(C) The two isomers cannot be separated by fractional distillation because both isomers have the same boiling point.
(D) The two isomers cannot be separated by fractional distillation because they interconvert rapidly at the temperatures at which they distill.

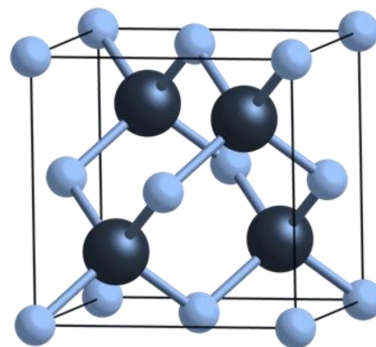
States Of Matter

56. A gas with $P = 615$ mm Hg is contained in the U-tube as shown. If $h = 65$ mm, what is the atmospheric pressure P_{atm} ?



- (A) 550 mm Hg (B) 615 mm Hg
(C) 680 mm Hg (D) 760 mm Hg
57. A beaker containing 25 mL of liquid 1-aminopentane, $\text{CH}_3(\text{CH}_2)_4\text{NH}_2$, is placed on a hotplate and brought to a boil. As the 1-aminopentane boils,
- (A) the total energy of the system stays constant.
(B) the hydrogen bonding between the 1-aminopentane molecules is disrupted.
(C) the ion-dipole forces between the 1-aminopentane molecules are disrupted.
(D) pentane and ammonia gas are formed.
58. Into both ends of a meter-long glass tube samples of gases are introduced simultaneously. One end receives hydrogen chloride gas (HCl) while the other end receives ammonia gas (NH_3). When the gases meet in the tube, they react to form solid ammonium chloride (NH_4Cl). Where in the tube does the NH_4Cl form?
- (A) At the center of the tube
(B) Closer to the end where the hydrogen chloride is inserted
(C) Closer to the end where the ammonia is inserted
(D) Uniformly at all positions in the tube
59. At its normal boiling point of -1.0°C , a sample of gaseous butane at 1.0 atm occupies a volume of 1.0 L. What is the pressure if the volume is decreased to 0.70 L while maintaining the temperature at -1.0°C ?
- (A) 0.70 atm (B) 1.0 atm
(C) 1.4 atm (D) 2.0 atm

60. What is the principal energetic factor in the lack of miscibility between $\text{C}_6\text{H}_{14}(\text{l})$ and $\text{H}_2\text{O}(\text{l})$?
- (A) The strength of intermolecular forces of attraction between $\text{C}_6\text{H}_{14}(\text{l})$ molecules
(B) The strength of intermolecular forces of attraction between $\text{H}_2\text{O}(\text{l})$ molecules
(C) The difference between the molecular weights of the molecules
(D) The difference in electronegativity between carbon and hydrogen
61. The melting point of silicon dioxide (1713°C) is higher than the melting point of silicon (1414°C). What is the best explanation for this difference?
- (A) Silicon-oxygen bonds are stronger than silicon-silicon bonds.
(B) Silicon dioxide is an ionic solid while silicon is a metallic solid.
(C) Silicon dioxide is polar while silicon is nonpolar.
(D) Silicon dioxide forms tetragonal crystals while silicon forms cubic crystals.
62. A unit cell of the cubic form of ZnS is shown below (large spheres = Zn, small spheres = S). How many of each type of atom are present in one unit cell?



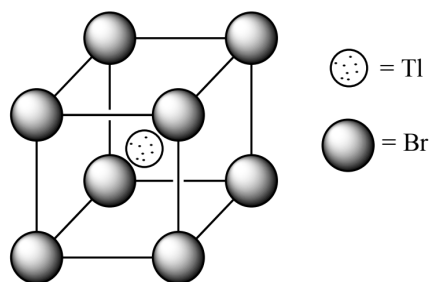
- (A) 1 Zn, 1 S (B) 2 Zn, 4 S
(C) 4 Zn, 4 S (D) 4 Zn, 14 S
63. Which compound has the highest normal boiling point?
- (A) $\text{CH}_3\text{CH}_2\text{COOH}$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
(C) $\text{CH}_3\text{COOCH}_3$ (D) $\text{HCOOCH}_2\text{CH}_3$

States Of Matter

64. Nitrogen, N_2 , has the following properties:
 normal melting point: 63.2 K
 normal boiling point: 77.4 K
 triple point: 0.127 atm, 63.1 K
 critical point: 33.5 atm, 126.0 K

Which statement about N_2 is correct?

- (A) Liquid N_2 is denser than solid N_2 .
 (B) At sufficiently high pressure, N_2 can be liquefied at 150 K.
 (C) Liquid N_2 and gaseous N_2 can coexist at 63.1 K and 1 atm.
 (D) If N_2 is heated from 60 K to 70 K at 0.100 atm, it sublimes.
65. The vapor pressure of iodomethane, CH_3I ($M = 141.9$), is 110. mm Hg at 266 K. A 0.824 g sample of iodomethane is placed in a closed, evacuated 370. mL container at 266 K. At equilibrium, what will be the pressure in the container?
- (A) 96.4 mm Hg (B) 110. mm Hg
 (C) 260. mm Hg (D) 292. mm Hg
66. Which statement about the properties of barium chloride and mercury(II) chloride is correct?
- (A) $BaCl_2$ has a higher melting point than $HgCl_2$.
 (B) $BaCl_2$ has a higher solubility in nonpolar solvents than $HgCl_2$.
 (C) $BaCl_2$ has a higher vapor pressure than $HgCl_2$.
 (D) Molten $BaCl_2$ has a lower electrical conductivity than molten $HgCl_2$.
67. Thallium(I) bromide ($M = 284.3$) adopts the cubic unit cell shown below, with an edge length $a = 397$ pm. What is the density of $TlBr$?



- (A) 5.42 g cm^{-3} (B) 7.55 g cm^{-3}
 (C) 18.0 g cm^{-3} (D) 22.4 g cm^{-3}

68. Which of the following properties is not typical of metallic solids?
- (A) High vapor pressure
 (B) High coordination number of atoms in the lattice
 (C) High electrical conductivity
 (D) High thermal conductivity

69. Which of the following substances experience London dispersion forces?

I. CH_3CH_3

II. CH_3OH

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

70. Which of these pure substances has the highest normal boiling point?

- (A) CH_4 (B) NH_3 (C) SiH_4 (D) PH_3

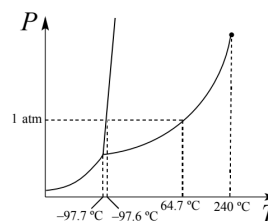
71. A mixture of 0.50 mol of H_2 gas and 1.3 mol of Ar gas is in a sealed container with a volume of 4.82 L. If the temperature of the mixture is 50.0°C , what is the partial pressure of H_2 in the sample?

- (A) 1.5 atm (B) 2.8 atm
 (C) 7.2 atm (D) 9.9 atm

72. For which pair of allotropes is one a molecular solid and the other a network covalent solid?

- (A) Dioxygen and ozone
 (B) White phosphorus and red phosphorus
 (C) Rhombic sulfur and monoclinic sulfur
 (D) Gray tin and white tin

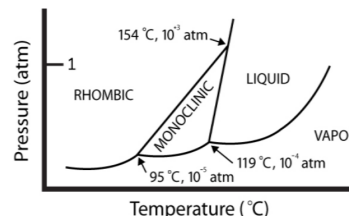
73. According to the phase diagram of methanol shown below, which statement is correct?



- (A) Solid methanol has a greater density than liquid methanol.
 (B) Solid methanol sublimates at atmospheric pressure.
 (C) Solid, liquid, and gaseous methanol can only coexist at pressures above 1 atm.
 (D) At 200°C and 1 atm pressure, methanol is a supercritical fluid.

States Of Matter

74. Which statement about atoms arranged in a body-centered cubic (bcc) crystal structure is correct?
- (A) It is not observed as the structure of any metallic elements.
 (B) It is also called the cubic closest-packed (ccp) structure.
 (C) The unit cell contains two atoms.
 (D) Each atom has 6 nearest neighbors.
75. Which physical property decreases with an increase in intermolecular molecular forces?
- (A) boiling point
 (B) enthalpy of vaporization
 (C) vapor pressure
 (D) viscosity
76. A 0.180 g sample of liquid H_2O is injected into a 5.00 L flask at 25°C . What will be present in the flask when equilibrium is established? (Vapor pressure H_2O at 25°C = 23.8 mm Hg)
- (A) H_2O vapor at a pressure of 186 mm Hg
 (B) H_2O vapor at a pressure of 37.2 mm Hg
 (C) liquid H_2O and H_2O vapor at a pressure of 37.2 mm Hg
 (D) liquid H_2O and H_2O vapor at a pressure of 23.8 mm Hg
77. The atmospheric pressure on the summit of Mt. Everest is 0.333 atmospheres. At what temperature (in $^\circ\text{C}$) does H_2O boil there? ($\Delta H_{\text{vap}} \text{H}_2\text{O} = 40.7 \text{ kJ}\cdot\text{mol}^{-1}$)
- (A) 71°C (B) 87°C (C) 96°C (D) 98°C
78. Benzene and toluene form an ideal solution. The vapor pressure of benzene at 55°C is 400 mm Hg while the vapor pressure of toluene at 55°C is 130 mm Hg. What is the vapor pressure of a solution consisting of 0.5 mole fraction of benzene and 0.5 mole fraction of toluene at 55°C ?
- (A) lower than 130 mm Hg
 (B) between 130 and 400 mm Hg
 (C) exactly 400 mm Hg
 (D) greater than 400 mm Hg
79. An oxide of rhenium crystallizes with eight rhenium atoms at the corners of the unit cell and 12 oxygen atoms on the edges between them. What is the formula of this oxide?
- (A) ReO (B) Re_2O_3 (C) ReO_2 (D) ReO_3
80. The Henry's law constant for oxygen gas in water at 25°C k_{O_2} is $1.3 \times 10^{-3} \text{ M}\cdot\text{atm}^{-1}$. What is the partial pressure of O_2 above a solution at 25°C with an O_2 concentration of $2.3 \times 10^{-4} \text{ M}$ at equilibrium?
- (A) 5.7 atm (B) 0.18 atm
 (C) $1.3 \times 10^{-3} \text{ atm}$ (D) $3.0 \times 10^{-7} \text{ atm}$
81. A sample of gas measured at 20°C and 4.0 atm is heated to 40°C at constant volume. Which statement(s) is (are) true of the gas after heating relative to its initial state?
- I. The average molecular kinetic energy is increased.
 II. The average molecular speed is unchanged.
 III. The pressure of the gas is increased to 8.0 atm.
 IV. The number of molecular collisions per second is unchanged.
- (A) I. only (B) I. and IV. only
 (C) II. and III. only (D) II. and IV. only
82. Under what conditions does the behavior of real gases deviate most from that predicted by the ideal gas law?
- (A) low P, low T (B) high P, low T
 (C) low P, high T (D) high P, high T
83. When 0.25 L of liquid nitrogen ($d = 0.807 \text{ g/mL}$) is vaporized, what volume does the resulting gas occupy at 25°C and 5.00 atm?
- (A) 71 L (B) 54 L (C) 35 L (D) 32 L
- 84.



The phase diagram for sulfur is shown above. Which statement about this diagram is correct?

- (A) The critical point is above 154°C and 10^3 atm .
 (B) There are only two triple points in this diagram.
 (C) Monoclinic sulfur is more dense than rhombic sulfur at any temperature.
 (D) Monoclinic sulfur forms rhombic sulfur at higher pressure or lower temperature.

States Of Matter

85. What is the principal difference between crystalline and amorphous solids?

crystalline solids amorphous solids

- (A) ionic bonding covalent bonding
(B) higher molar masses lower molar masses
(C) stoichiometric solids non-stoichiometric solids
(D) long-range order lack of long-range order

86. A sample of H_2 collected over H_2O at $23^\circ C$ and a pressure of 732 mm Hg has a volume of 245 mL. What volume would the dry H_2 occupy at $0^\circ C$ and 1 atm pressure?

[vp H_2O at $23^\circ C$ = 21 mm Hg]

- (A) 211 mL (B) 218 mL
(C) 224 mL (D) 249 mL

87. Two samples of gas, one of argon and one of helium, have the same pressure, temperature and volume. Which statement is true assuming both gases behave ideally?

- (A) The helium sample contains more atoms than the argon sample and the helium atoms have a higher average speed.
(B) The two samples have the same number of atoms but the helium atoms have a higher average speed.
(C) The two samples have the same number of atoms and both types of atoms have the same average speed.
(D) The two samples have the same number of atoms but the argon atoms have a higher average speed.

88. For a sample of liquid in a closed container, which aspect(s) of vaporization depend on the surface area of the liquid?

I rate of vaporization II vapor pressure

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

89. The formulas and boiling points of three compounds are given in this table.

Formula	$CH_3CH_2CH_3$	CH_3OCH_3	CH_3CHO
BP, K	231	250	294

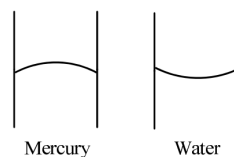
The trend in boiling points is best attributed to variations in

- (A) covalent bonding. (B) dipole forces.
(C) dispersion forces. (D) hydrogen bonding.

90. Which statement about the triple point of a substance is correct?

- (A) The triple point for a substance varies with the pressure.
(B) The three phases (solid, liquid, gas) have the same density.
(C) The three phases (solid, liquid, gas) are in equilibrium.
(D) The three phases (solid, liquid, gas) are indistinguishable in appearance.

91. In a glass tube columns of water and mercury appear as shown.



This is best attributed to the differences in their

- (A) densities. (B) molar masses.
(C) surface tensions. (D) viscosities.

92. Compounds of uranium-235 and uranium-238 can be separated from one another by

- (A) distillation.
(B) effusion.
(C) fractional crystallization.
(D) paper chromatography.

93. A 5.00 L evacuated cylinder is charged with 25.5 g of NH_3 and 36.5 g of HCl . Calculate the final pressure at $85.0^\circ C$ after the two compounds have reacted completely. $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$

- (A) 2.94 atm (B) 5.88 atm
(C) 8.82 atm (D) 14.7 atm

94. Pure samples of which of the following exhibit hydrogen bonding?

I CH_3OH II CH_3NO_2 III CH_3CN

- (A) I only (B) I and II only
(C) II and III only (D) I, II, and III

95. The average molecular velocity in a gas sample at 300 K is 500 m/s. The temperature of this gas is increased until the average velocity of its molecules is 1000 m/s. What is the new temperature?

- (A) 420 K (B) 573 K (C) 600 K (D) 1200 K

States Of Matter

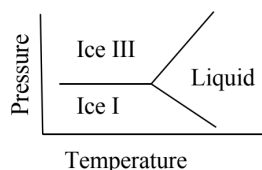
96. Which of the following compounds has the lowest boiling point?

(A) HF (B) HCl (C) HBr (D) HI

97. What is the coordination number of each atom in a hexagonal close-packed solid?

(A) 4 (B) 6 (C) 8 (D) 12

98. Water ice exists in several different forms depending on the pressure and temperature. A portion of the phase diagram for ice I, ice III, and liquid water is shown below.



Which statement about the densities of these three phases is correct?

- (A) The density of liquid water is greater than the densities of either ice I or ice III.
 (B) The density of ice I is greater than the densities of either ice III or liquid water.
 (C) The density of ice III is greater than the densities of either ice I or liquid water.
 (D) The densities of ice I and ice III are equal and greater than the density of liquid water.

99. Ar and He are both gases at room temperature. How do the average molecular velocities (V) of their atoms compare at this temperature?

(A) $V_{\text{He}} = 10V_{\text{Ar}}$ (B) $V_{\text{Ar}} = 10V_{\text{He}}$
 (C) $V_{\text{He}} = 3V_{\text{Ar}}$ (D) $V_{\text{Ar}} = 3V_{\text{He}}$

100. Lithium reacts with water to produce hydrogen gas and lithium hydroxide. What volume of hydrogen collected over water at 22°C and 750 mm Hg pressure is produced by the reaction of 0.208 g of Li? [$VP_{\text{H}_2\text{O}} = 19.8 \text{ mm Hg}$]

(A) 367 mL (B) 378 mL
 (C) 735 mL (D) 755 mL

101. Correct statements about samples of ice and liquid water at 0°C include which of the following?

I Molecules in ice and liquid water have the same kinetic energy.
 II Liquid water has a greater entropy than ice.
 III Liquid water has a greater potential energy than ice.

(A) I and II only (B) I and III only
 (C) II and III only (D) I, II, and III

102. A sample of a volatile liquid is introduced to an evacuated container with a movable piston. Which change occurs as the piston is raised? (Assume some liquid remains.)

I The fraction of the molecules in the gas phase increases
 II The pressure in the container decreases

(A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

103. The kinetic energy of the molecules in a sample of H_2O in its stable state at -10°C and 1 atm is doubled. What are the initial and final phases?

(A) solid $\rightarrow$ liquid (B) liquid $\rightarrow$ gas
 (C) solid $\rightarrow$ gas (D) solid $\rightarrow$ solid

104. Barium metal crystallizes in a body-centered cubic lattice with barium atoms only at the lattice points. If the density of barium metal is 3.50 g/cm^3 , what is the length of the unit cell?

(A) $3.19 \times 10^{-8} \text{ cm}$ (B) $4.02 \times 10^{-8} \text{ cm}$
 (C) $5.07 \times 10^{-8} \text{ cm}$ (D) $6.39 \times 10^{-8} \text{ cm}$

105. Which statement is not a principle (postulate) of kinetic molecular theory?

(A) The molecules of a gas are in rapid random motion.
 (B) The molecules of an ideal gas exhibit no attractive forces.
 (C) The collisions of gaseous molecules with one another and the walls of their container are elastic.
 (D) Equal volumes of gases at the same temperature and pressure contain equal numbers of molecules.

106. When a sample of an ideal gas is heated from 25°C to 50°C the average kinetic energy of the molecules increases. Which ratio gives the correct relationship between the average kinetic energies at the higher temperature to the lower temperature?

(A) 2 : 1 (B) $\sqrt{2} : \sqrt{1}$
 (C) 323 : 298 (D) $\sqrt{323} : \sqrt{298}$

107. A partially filled tank of propane contains both a liquid and a gas phase. Which of these statements about the contents of the two phases are correct?

I. The two phases have the same potential energy but different kinetic energies.
 II. The two phases have the same molar masses but different densities.

(A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

States Of Matter

108. Which property or properties of metals can be accounted for by the electron sea model?

I. electrical conductivity
II. malleability

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
109. In a crystal of a typical metallic element, an atom has how many nearest neighbors?
- (A) 4 (B) 6 (C) 12 (D) 16
110. Which substance has the highest melting point?
- (A) Li_2O (B) MgO (C) CO_2 (D) N_2O_5

111. Which solute is least soluble in water?

- (A) 1-butanol (B) ethanol
(C) methanol (D) 1-propanol

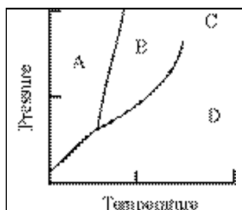
112. For which two gases are the rates of effusion 2:1?

- (A) H_2 and He (B) He and O_2
(C) Ne and Kr (D) N_2 and Ar

113. Which gas has a density of $0.71 \text{ g}\cdot\text{L}^{-1}$ at 0°C and 1 atm?

- (A) Ar (B) Ne (C) CO (D) CH_4

114. Supercritical carbon dioxide exists at which point on the accompanying phase diagram?



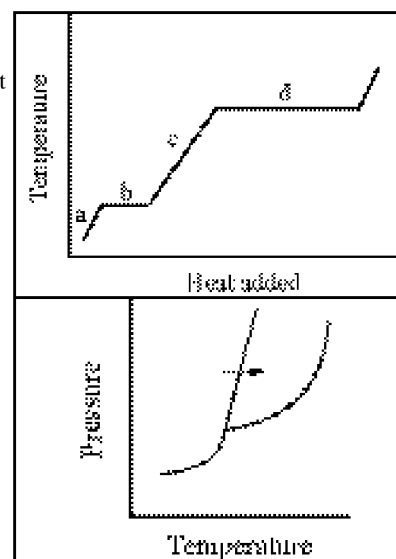
- (A) A (B) B (C) C (D) D

115. Which properties increase with an increase in intermolecular forces at 25°C ?

I. surface tension
II. vapor pressure

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
116. The atoms in crystals of silver metal are arranged in a cubic closest packed structure. What is the unit cell in this structure?
- (A) body-centered cubic (B) face-centered cubic
(C) hexagonal-close packed (D) simple cubic

117. Which segment of the heating curve obtained at constant pressure corresponds to the transition denoted by the arrow in the phase diagram?



- (A) a (B) b (C) c (D) d

118. What is the molar mass of a gas that has a density of $5.66 \text{ g}\cdot\text{L}^{-1}$ at 35°C and 745 mm Hg?

- (A) 127 (B) 141 (C) 143 (D) 146

119. Consider the solids: body-centered cubic (bcc), face-centered cubic (fcc), simple cubic (sc) (or primitive), constructed of spheres of the same size. When they are arranged in increasing order of the percentage of free space in a unit cell, which order is correct?

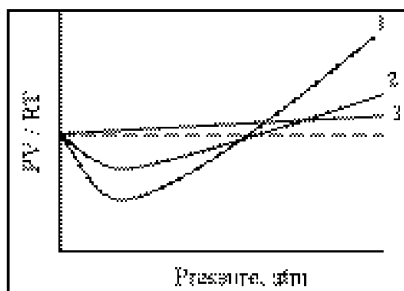
- (A) fcc, bcc, sc (B) bcc, sc, fcc
(C) sc, fcc, bcc (D) bcc, fcc, sc

120. If the absolute temperature of a sample of gas is increased by a factor of 1.5, by what ratio does the average molecular speed of its molecules increase?

- (A) 1.2 (B) 1.5 (C) 2.2 (D) 3.0

States Of Matter

121. The curves in the accompanying diagram represent the PV/RT behavior of the gases: He, CH₄ and C₃H₈. Which assignment of behavior to gas is correct?



- (A) 1 = He
2 = CH₄
3 = C₃H₈
- (B) 1 = C₃H₈
2 = CH₄
3 = He
- (C) 1 = CH₄
2 = C₃H₈
3 = He
- (D) 1 = C₃H₈
2 = He
3 = CH₄
122. Solid calcium occurs as either cubic closest packing or hexagonal closest packing. What is the most significant difference between these two structures?
- (A) the placement of layers of calcium atoms
- (B) the distance between calcium atoms in a single layer
- (C) the distance between calcium atoms in adjacent layers
- (D) the coordination number of the calcium atoms in a single layer

123. When the substances are arranged in order of increasing boiling points, which order is correct?

I. CH₃(CH₂)₃CH₃
II. CH₃CH₂CH(CH₃)₂
III. C(CH₃)₄

- (A) I < II < III
- (B) II < III < I
- (C) III < II < I
- (D) III < I < II

124. A 225 mL sample of H₂ is collected over water at 25 °C and 735 mmHg pressure. Which expression represents the set-up to find the volume of dry H₂ at 0 °C and 1 atmosphere?

Vapor Pressure at 25 °C	
H ₂ O	24 mmHg

- (A) $V = \frac{225 \times (735 - 24) \times 273}{760 \times 298}$
- (B) $V = \frac{225 \times 760 \times 298}{(735 - 24) \times 273}$
- (C) $V = \frac{225 \times 273 \times 760}{(735 + 24) \times 298}$
- (D) $V = \frac{225 \times (735 + 24) \times 298}{760 \times 273}$

125. What occurs when liquid CH₂F₂ evaporates?

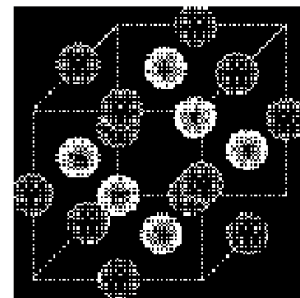
I. Dispersion forces are overcome.
II. Dipole-dipole forces are overcome.
III. Covalent bonds are broken.

- (A) II only
- (B) III only
- (C) I and II only
- (D) I, II and III

126. In the van der Waals equation for real gases, corrections are introduced for both the pressure and the volume terms of the Ideal Gas Equation. Identify the origin of both correction factors and specify whether each is added to or subtracted from the corresponding term.

Pressure	Volume
(A) attractive forces / subtracted	molecular size / added
(B) attractive forces / added	molecular size / subtracted
(C) molecular size / subtracted	attractive forces / added
(D) molecular size / added	attractive forces / subtracted

127. The structure of a unit cell of an oxide of niobium is depicted here. Niobiums are dark and oxygens are light. What is the empirical formula of this compound?



- (A) NbO
- (B) NbO₂
- (C) NbO₃
- (D) Nb₂O₃

128. The solubility of a gas in a liquid increases when which of the following increases?

I. pressure of the gas
II. temperature of the liquid

- (A) I only
- (B) II only
- (C) both I and II
- (D) neither I nor II

129. Which property is the same for 1.0 g samples of H₂ and CH₄ in separate 1.0 L containers at 25 °C?

- (A) pressure
- (B) number of molecules
- (C) average molecular velocity
- (D) average molecular kinetic energy

States Of Matter

130. When CsI, SiO₂, CH₃OH and C₃H₈ are listed in order of increasing melting point, which is the correct order?

(A) CsI, SiO₂, CH₃OH, C₃H₈
 (B) CH₃OH, C₃H₈, CsI, SiO₂
 (C) CH₃OH, C₃H₈, SiO₂, CsI
 (D) C₃H₈, CH₃OH, CsI, SiO₂

131. An unknown gas effuses through a pin-hole in a container at a rate of 7.2 mmol/s. Under the same conditions gaseous oxygen effuses at a rate of 5.1 mmol/s. What is the molar mass (in g/mol) of the unknown gas?

(A) 16 (B) 23 (C) 45 (D) 64

132. When NaF, MgO, KCl and CaS are listed in order of increasing lattice energy, which order is correct?

(A) MgO, NaF, KCl, CaS (B) CaS, MgO, KCl, NaF
 (C) KCl, CaS, NaF, MgO (D) KCl, NaF, CaS, MgO

133. When compared to most other substances of similar molar mass the values of which properties of liquid H₂O are unusually large?

I. boiling point
 II. specific heat capacity
 III. surface tension

(A) I only (B) I and II only
 (C) II and III only (D) I, II and III

134. An unknown gas is placed in a sealed container with a fixed volume. Which of the characteristics listed change(s) when the container is heated from 25 °C to 250 °C?

I The density of the gas
 II The average kinetic energy of the molecules
 III The mean free path between molecular collisions

(A) I only (B) II only
 (C) III only (D) I and II only

135. Which gas has the same density at 546 °C and 1.50 atm as that of O₂ gas at STP?

(A) N₂ (B) NH₃ (C) SO₂ (D) SO₃

136. Which plot involving vapor pressure (VP) and absolute temperature results in a straight line?

(A) VP vs T (B) VP vs T⁻¹
 (C) ln VP vs T (D) ln VP vs T⁻¹

137. What is the order of the boiling points (from lowest to highest) for the hydrogen halides?

(A) HF < HCl < HBr < HI (B) HI < HBr < HCl < HF
 (C) HCl < HF < HBr < HI (D) HCl < HBr < HI < HF

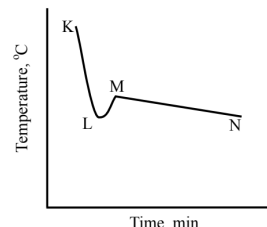
138. Of the three types of cubic lattices, which have the highest and lowest densities for the same atoms?

Highest

Lowest

(A) simple cubic body-centered cubic
 (B) face-centered cubic simple cubic
 (C) body-centered cubic face-centered cubic
 (D) face-centered cubic body-centered cubic

139. The curve shown results when a liquid is cooled. What temperature is closest to the freezing point of the liquid?



(A) L (B) M
 (C) $\frac{L+M}{2}$ (D) $\frac{M+N}{2}$

140. Which change increases the solubility of a gas in water?

1. an increase in water temperature
 2. a decrease in gas pressure

(A) 1 only (B) 2 only
 (C) both 1 and 2 (D) neither 1 nor 2

141. A 200. mL sample of a gaseous hydrocarbon has a density of 2.53 g·L⁻¹ at 55 °C and 720 mmHg. What is its formula?

(A) C₂H₆ (B) C₄H₁₀ (C) C₅H₁₂ (D) C₆H₆

142. Sulfur and fluorine form SF₆ and S₂F₁₀, both of which are gases at 30 °C. When an equimolar mixture of them is allowed to effuse through a pinhole, what is the ratio SF₆/S₂F₁₀ in the first sample that escapes?

Molar Mass g·mol ⁻¹	
SF ₆	146
S ₂ F ₁₀	254

(A) 1.32/1 (B) 1.74/1 (C) 3.03/1 (D) 3.48/1

143. The volumes of real gases often exceed those calculated by the ideal gas equation. These deviations are best attributed to the

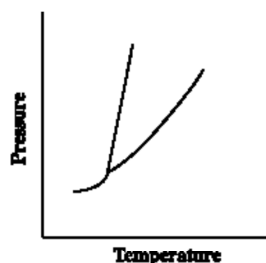
(A) attractive forces between the molecules in real gases.
 (B) dissociation of the molecules in real gases.
 (C) kinetic energy of the molecules in real gases.
 (D) volumes of the molecules in real gases.

States Of Matter

144. The electrical conductivity of a solid is slight at 25 °C and much greater at 125 °C. The solid is most likely a(n)

(A) ionic compound (B) insulator
(C) metal (D) semiconductor

145. According to the phase diagram, what would be the effect of increasing the pressure on this substance?

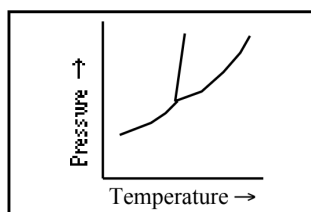


(A) decrease both the melting and boiling points
(B) increase both the melting and boiling points
(C) increase the melting point and decrease the boiling point
(D) decrease the melting point and increase the boiling point

146. Equal masses of gaseous N_2 , NH_3 , and N_2O are injected into an evacuated container to produce a total pressure of 3 atm. How do the partial pressures of N_2 , NH_3 , and N_2O compare?

(A) $P_{N_2} = P_{NH_3} = P_{N_2O}$ (B) $P_{N_2} < P_{NH_3} < P_{N_2O}$
(C) $P_{NH_3} < P_{N_2} < P_{N_2O}$ (D) $P_{N_2O} < P_{N_2} < P_{NH_3}$

147. According to this phase diagram, which phases can exist at pressures lower than the triple point pressure?



(A) gas only (B) solid and gas only
(C) liquid only (D) solid and liquid only

148. 1.00 g of water is introduced into a 5.00 L evacuated flask at 50 °C. What mass of water is present as liquid when equilibrium is established?

Vapor Pressure, 50 °C	
H_2O	92.5 mmHg

(A) 0.083 g (B) 0.41 g
(C) 0.59 g (D) 0.91 g

149. Which substance has the greatest lattice energy?

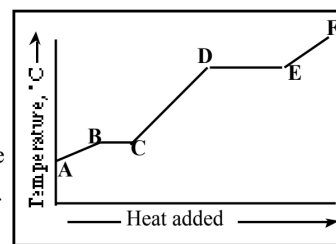
(A) NaF (B) KCl (C) MgO (D) CaS

150. When the temperature of a sample of H_2S gas is lowered, the pressure decreases more than predicted by the ideal gas equation. To what is this deviation from expected behavior due?

1. attractive forces between molecules
2. mass of the molecules
3. volume of the molecules

(A) 1 only (B) 2 only
(C) 1 and 3 only (D) 2 and 3 only

151. This curve is produced when a pure substance is heated. Which characteristic of this curve is related to the value for the enthalpy of fusion of the substance?



(A) length of AB (B) length of BC
(C) slope of AB (D) slope of CD

152. What is the most likely boiling point of an equimolar mixture of hexane, C_6H_{14} , and heptane, C_7H_{16} ?

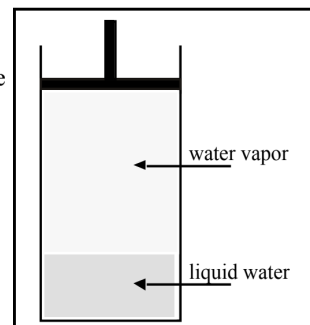
Boiling Point	
C_6H_{14}	69 °C
C_7H_{16}	98 °C

(A) below 69 °C (B) between 69 and 98 °C
(C) 69 °C (D) 98 °C

153. The triple point of CO_2 occurs at 5.1 atm and -56 °C. Its critical temperature is 31 °C. Solid CO_2 is more dense than liquid CO_2 . Under which combination of pressure and temperature is liquid CO_2 stable at equilibrium?

(A) 10 atm and -25 °C (B) 5.1 atm and -25 °C
(C) 10 atm and 33 °C (D) 5.1 atm and -100 °C

154. The vapor pressure of water at 20 °C is 17.54 mmHg. What will be the vapor pressure of the water in the apparatus shown after the piston is lowered, decreasing the volume of the gas above the liquid to one half of its initial volume? (Assume no temperature change.)



(A) 8.77 mmHg
(B) 17.54 mmHg
(C) 35.08 mmHg
(D) between 8.77 and 17.54 mmHg

States Of Matter

155. What is the density of propane, C_3H_8 , at 25°C and 740. mmHg?

(A) $0.509\text{ g}\cdot\text{L}^{-1}$ (B) $0.570\text{ g}\cdot\text{L}^{-1}$
(C) $1.75\text{ g}\cdot\text{L}^{-1}$ (D) $1.96\text{ g}\cdot\text{L}^{-1}$

156. An unknown gas effuses through a small hole one half as fast as methane, CH_4 , under the same conditions. What is the molar mass of the unknown gas?

(A) $4\text{ g}\cdot\text{mol}^{-1}$ (B) $8\text{ g}\cdot\text{mol}^{-1}$
(C) $32\text{ g}\cdot\text{mol}^{-1}$ (D) $64\text{ g}\cdot\text{mol}^{-1}$

157. What is the average velocity of H_2 molecules at 100 K relative to their velocity at 50 K?

(A) 2.00 times the velocity at 50 K
(B) 1.41 times the velocity at 50 K
(C) 0.71 times the velocity at 50 K
(D) 0.50 times the velocity at 50 K

158. The heat of formation of NO from its elements is $+90\text{ kJ}\cdot\text{mol}^{-1}$. What is the approximate bond dissociation energy of the bond in NO?

Bond	Bond Energy
$\text{N}\equiv\text{N}$	$941\text{ kJ}\cdot\text{mol}^{-1}$
$\text{O}=\text{O}$	$499\text{ kJ}\cdot\text{mol}^{-1}$

(A) $630\text{ kJ}\cdot\text{mol}^{-1}$ (B) $720\text{ kJ}\cdot\text{mol}^{-1}$
(C) $765\text{ kJ}\cdot\text{mol}^{-1}$ (D) $810\text{ kJ}\cdot\text{mol}^{-1}$

159. What is the total pressure in a 2.00 L container that holds 1.00 g He, 14.0 g CO, and 10.0 g of NO at 27.0°C ?

(A) 21.6 atm (B) 13.2 atm
(C) 1.24 atm (D) 0.310 atm

160. What type of solid is generally characterized by having low melting point and low electrical conductivity?

(A) ionic (B) metallic
(C) molecular (D) network covalent

161. How many nearest neighbors surround each particle in a face-centered cubic lattice?

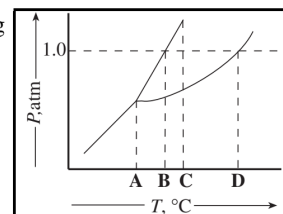
(A) 4 (B) 6 (C) 8 (D) 12

162. Hydrogen is collected over water at 22°C and a barometer reading of 740 mmHg. If 300. mL of hydrogen is collected, which expression will give the volume of dry hydrogen at the same temperature and pressure?

Compound	Vapor Pressure at 22°C
H_2O	20. mmHg

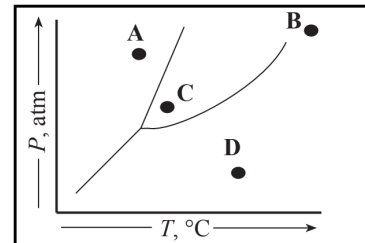
(A) $300\text{ mL} \times \frac{740\text{ mmHg} - 20\text{ mmHg}}{740\text{ mmHg}}$
(B) $300\text{ mL} \times \frac{740\text{ mmHg} + 20\text{ mmHg}}{740\text{ mmHg}}$
(C) $300\text{ mL} \times \frac{740\text{ mmHg}}{740\text{ mmHg} - 20\text{ mmHg}}$
(D) $300\text{ mL} \times \frac{740\text{ mmHg}}{740\text{ mmHg} + 20\text{ mmHg}}$

163. What is the normal melting point of the substance represented by the phase diagram?



(A) A (B) B (C) C (D) D

164. Which point in the phase diagram best represents supercritical conditions?



(A) A (B) B (C) C (D) D

165. The vapor pressure of a liquid in a closed container depends on

1. temperature of the liquid
2. quantity of liquid
3. surface area of the liquid

(A) 1 only (B) 2 only
(C) 1 and 3 only (D) 1, 2 and 3

166. What is the maximum number of phases that can be in equilibrium in a one component system?

(A) 1 (B) 2 (C) 3 (D) 4

167. The molar mass of a gas with a density of $5.8\text{ g}\cdot\text{L}^{-1}$ at 25°C and 740 mm Hg is closest to

(A) $10\text{ g}\cdot\text{mol}^{-1}$ (B) $20\text{ g}\cdot\text{mol}^{-1}$
(C) $150\text{ g}\cdot\text{mol}^{-1}$ (D) $190\text{ g}\cdot\text{mol}^{-1}$

States Of Matter

168. Which substance would be expected to exhibit the greatest surface tension at 25 °C?
- (A) CH_3OCH_3 (B) $\text{C}_2\text{H}_5\text{OH}$
(C) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ (D) $\text{CH}_2(\text{OH})\text{CH}_2\text{OH}$